University of Belgrade, Faculty of Sport and Physical Education

In-class work — week 2

Statistics in Physical Education and Sport · chapters 6 and 7 · jamovi, about 45 minutes

The data are in two files: STAT_DAS_N02_Class_Height_mass.xlsx (height in meters and mass in kilograms of 11-year-old boys and girls, N = 1982) and STAT_DAS_N02_Class_Jumps.xlsx (three jumps, ten children at each age: 12, 13 and 14 years). For each step, write down the requested number, table or sentence. Save your answers, jamovi tables and plots in one document and submit it in Teams (assignment “02. In-class work”) before the next class. If you do not reach part C in class, finish it at home.

A. The distribution of height and mass (chapter 6)

1. Set the measure type of sex to Nominal. Exploration → Descriptives: height_m and mass_kg in Variables, sex in Split by; under Statistics select N, Mean, Median, Std. deviation, Minimum, Maximum, Skewness, Kurtosis and Shapiro-Wilk. Copy the results into one summary table: rows are the variables by sex, columns the statistics.
2. Are height and mass normally distributed? For each variable and sex, compute the z value for skewness (Skewness divided by Std. error skewness) and for kurtosis, then look at the histogram and the Q-Q plot (Plots). Answer in two or three sentences: which variable departs more from the normal distribution, and in which direction? Why, with almost a thousand children, is the Shapiro-Wilk p value small even when the departure is very small?
3. Find the z scores for height and mass of the 500th boy (ID 500) and the 500th girl (ID 1554). In jamovi: Data → Compute, a new variable z_height with the formula Z(height_m, group_by = sex), and the same for mass. Check one value by hand, $z = (x - M) \div SD$, using the mean and standard deviation of that child's sex. Write down the four z scores, and in one sentence what the 500th boy's z score for height means.
4. Find the deciles of height and mass separately for boys and girls: Descriptives → Statistics → Percentile Values → Cut points for 10 equal groups. Write down the table of deciles. Between which deciles is the 500th boy's height?

B. From the sample to the population (chapter 7)

5. In Descriptives, select Std. error of Mean and Confidence interval for Mean (95%). Write down the standard error and the confidence interval for the mean height of boys and girls. Compute the interval by hand too, $M \pm 1.96 \cdot SE$, and compare it with jamovi. Why are they almost the same, and why would they not be with a sample of ten? (Lecture part 2: jamovi uses the t distribution with N − 1 degrees of freedom.)
6. Do 11-year-old boys and girls differ in height? Write the null and the (nondirectional) alternate hypothesis, and set $\alpha = .05$ before computing anything.
7. Compute the mean difference (girls − boys) and the standard error of the difference, $SE = \sqrt{(SE_1^2 + SE_2^2)}$. Compute z = difference ÷ SE, the two-tailed p value (jamovi, distrACTION module → Normal Distribution), and the 95% confidence interval of the difference. Write down the decision about the null hypothesis, and in one sentence check whether the interval leads to the same decision.
8. Check your work in jamovi: T-Tests → Independent Samples T-Test, height_m in Dependent Variables, sex in Grouping Variable; select Welch's, Mean difference and Confidence interval. We cover this test in detail in chapter 10; with samples this large, the t value is almost the same as your z value. Write down the t value, the p value and the interval.

9. Imagine that someone, only after looking at the means, stated the directional hypothesis “girls are taller”. What would the p value be, and what decision would they make? Explain in two sentences why this is not allowed, and what their actual significance level (α) is (lecture part 4).

C. Jumps by age

10. Open STAT_DAS_N02_Class_Jumps.xlsx and set age to Nominal. Descriptives for all three jumps, age in Split by: N, Mean, Std. deviation, Minimum, Maximum. Put the results in one table.
11. Plot the means by age for each jump: ANOVA → One-Way ANOVA, the jumps in Dependent Variables, age in Grouping Variable, Additional Statistics → Descriptives plots. Use only the plot; the test itself belongs to chapter 11. What do the lines around each point show? (Lecture part 2.)
12. In a word processor (for example Word), write a short report: the table from step 10, the three plots from step 11, and a comment of three to five sentences. How do the jumps change with age? What do samples of ten children per age say about how certain that conclusion is?

References

Weir, J. P., & Vincent, W. J. (2021). *Statistics in kinesiology* (5th ed.). Human Kinetics.

Mayer, B., Rihs, M., & Redsell-Montgomerie, C. (2026). *distrACTION: Quantiles and probabilities of continuous and discrete distributions* (Version 1.2.2) [jamovi module]. https://github.com/methodenlehre/distrACTION

The jamovi project. (2026). *jamovi* (Version 2.7) [Computer software]. https://www.jamovi.org

Use of artificial intelligence: the author led the work on this material from the initial idea to the finished content, designing it, setting the rules for writing and language use, and checking every part. Claude Code (Anthropic), an AI-based assistant, was used to write the programs that generate the slides, figures and documents, to draft text, and to check the language against the author's rules. The author is responsible for the content.